# Hierarchical Cancer Classification Report

*AI-assisted decision support summary for brain_tumor_0002.jpg*

## Executive Summary

| | | | |
|---|---|---|---|
| **Date & Time** | 2026-04-19 16:53:32 | **Device** | cpu |
| **Organ Model** | Ready (resnet50_organ_classifier.pth) | **Subtype Model** | Ready (resnet50_subtype_classifier_best.pth) |

### Final Decision

## Pituitary Tumor

**Outcome: ABNORMAL** Confidence: 97.01%

Label: Pituitary Tumor Pipeline Status: HIGH_CONFIDENCE

Tissue Route: Brain Tissue Subtype / Outcome Detail: Pituitary Tumor

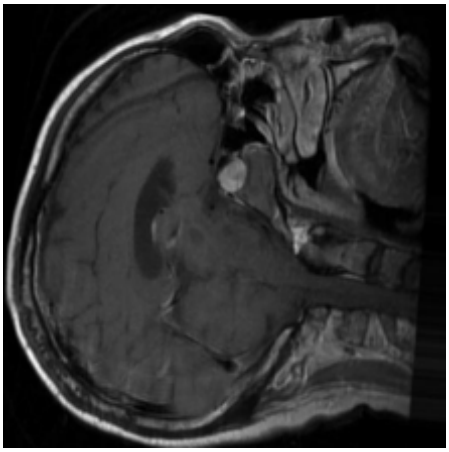
AI-generated decision support summary prepared for review.

## Visual Review

# Hierarchical Cancer Classification Report

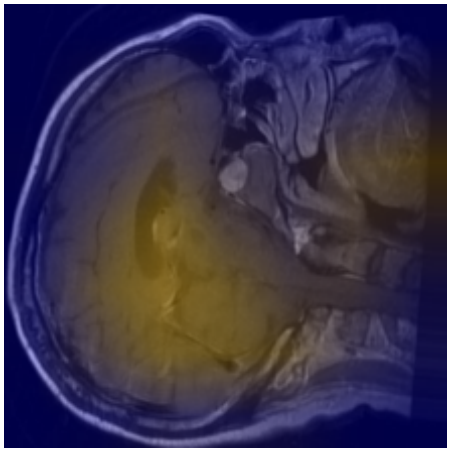
*AI-assisted decision support summary for brain_tumor_0002.jpg*

## Input Scan



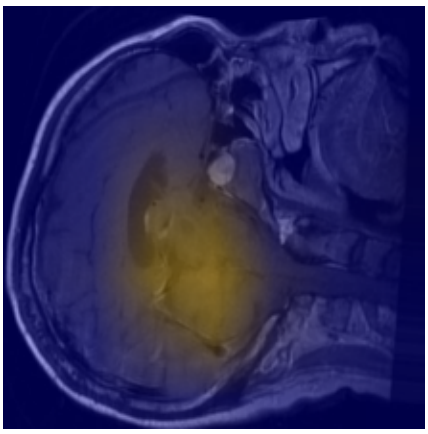
*Submitted image used for analysis.*

## Grad-CAM: Organ Routing



*Brain Tissue*

## Grad-CAM: Final Outcome



*Pituitary Tumor*

## Clinical Routing Summary

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*AI-assisted decision support summary for brain_tumor_0002.jpg*

| | |
|---|---|
| **Outcome** | ABNORMAL |
| **Confidence** | 97.01% |
| **Entropy** | 0.16 |
| **Label** | Healthy Brain Tissue |
| **Reason** | None |

## Organ / Tissue Probabilities

| Organ | Probability (%) |
|---|---|
| Brain Tissue | 99.71% |
| Cervical Tissue | 0.09% |
| Kidney Tissue | 0.05% |
| Lung & Colon Tissue | 0.04% |
| Breast Tissue | 0.04% |
| Lymphatic System | 0.03% |
| Oral Tissue | 0.03% |
| Blood (Leukemia-related) | 0.01% |

## Subtype Probabilities

| Subtype | Probability (%) |
|---|---|
| Pituitary Tumor | 97.01% |
| Healthy Brain Tissue | 2.17% |
| Brain Meningioma | 0.60% |
| Brain Glioma | 0.21% |

## Warnings / Notes

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- Relaxed validation applied for MRI.